

ASSIGNMENT NO.4: Working with Multiple Tables and Joins

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Batch	SY. BSc Data Science	Topic	WB(Assignment-4)

Q1) Create a MySQL database named CollegeDB and create the following tables:

Student Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
Column Name	Data Type	Constraint
City	VARCHAR(30)	

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	
Fees	DECIMAL(10,2)	

Enrollment Table

Column Name	Data Type	Constraint
EnrollmentID	INT	PRIMARY KEY
StudentID	INT	FOREIGN KEY
CourseID	INT	FOREIGN KEY
EnrollmentDate	DATE	

QUERY:

```
CREATE DATABASE SeniorCollegeDB;

USE SeniorCollegeDB;

CREATE TABLE Student (
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(50),
    City VARCHAR(30)
);

CREATE TABLE Course (
    CourseID INT PRIMARY KEY,
    CourseName VARCHAR(50),
    Fees DECIMAL(10,2)
);

CREATE TABLE Enrollment (
    EnrollmentID INT PRIMARY KEY,
    StudentID INT,
    CourseID INT,
    EnrollmentDate DATE,
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
    FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);
```

OUTPUT:

localhost / 127.0.0.1 | phpMyAdmin

localhost/phpmyadmin/index.php?route=/server/sql

Server: 127.0.0.1

Databases SQL Status User accounts Export Import Settings Replication Variables Charsets Engines Plug-ins

Run SQL query/queries on server "127.0.0.1":

```
1 CREATE DATABASE SeniorCollegeDB;
2
3 USE SeniorCollegeDB;
4
5 CREATE TABLE Student (
6   StudentID INT PRIMARY KEY,
7   StudentName VARCHAR(50),
8   City VARCHAR(30)
9 );
10
11 CREATE TABLE Course (
12   CourseID INT PRIMARY KEY,
13   CourseName VARCHAR(50),
14   Fees DECIMAL(10,2)
15 );
16
17 CREATE TABLE Enrollment (
18   EnrollmentID INT PRIMARY KEY,
19   StudentID INT,
20   CourseID INT,
21   EnrollmentDate DATE,
22   FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
23   FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
24 );
```

Clear Format Get auto-saved query

☐ Bind parameters

Console

localhost / 127.0.0.1 | phpMyAdmin

localhost/phpmyadmin/index.php?route=/server/sql

Server: 127.0.0.1

Databases SQL Status User accounts Export Import Settings Replication Variables Charsets Engines Plug-ins

Hide query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0013 seconds.)

```
CREATE DATABASE SeniorCollegeDB;
```

[Edit inline] [Edit] [Create PHP code]

⚠ Error: #1046 No database selected

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0002 seconds.)

```
USE SeniorCollegeDB;
```

[Edit inline] [Edit] [Create PHP code]

⚠ Error: #1046 No database selected

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0208 seconds.)

```
CREATE TABLE Student ( StudentID INT PRIMARY KEY, StudentName VARCHAR(50), City VARCHAR(30) );
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0180 seconds.)

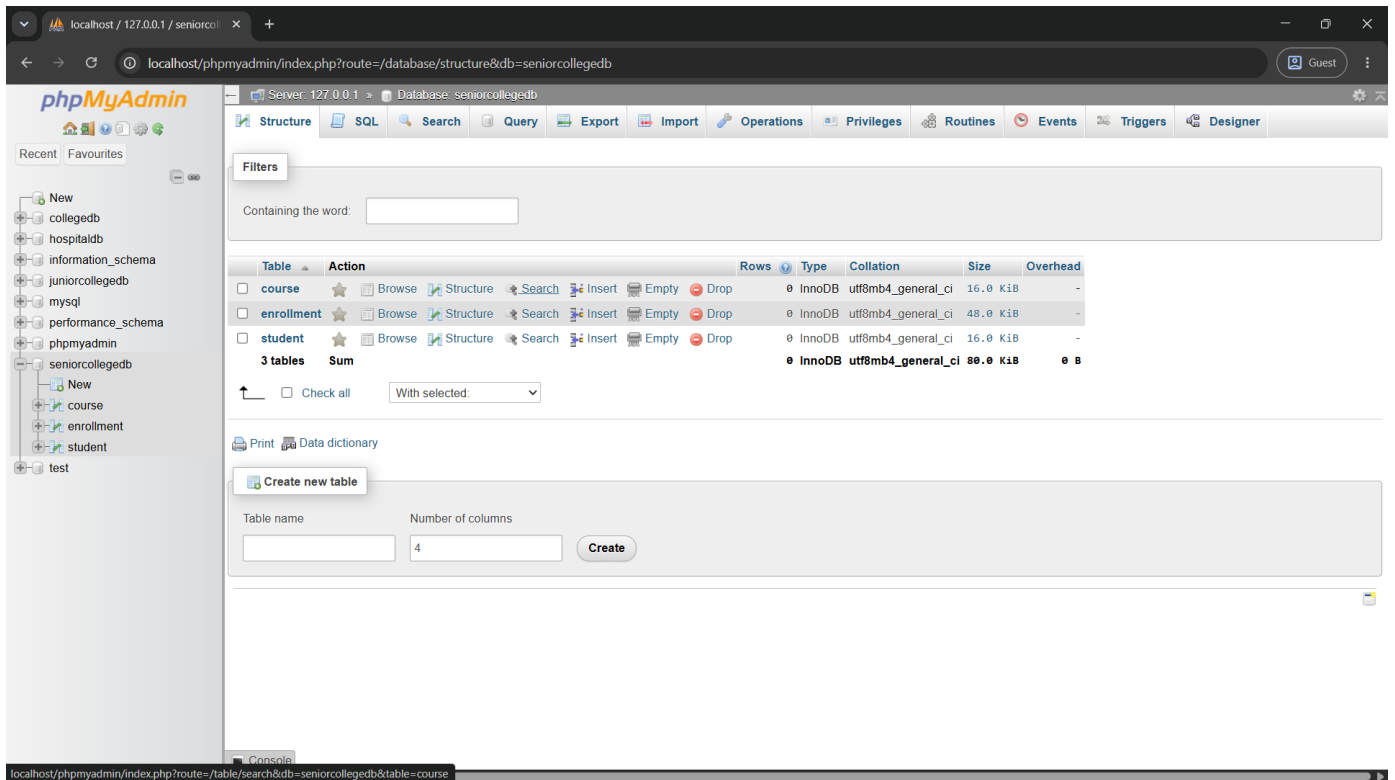
```
CREATE TABLE Course ( CourseID INT PRIMARY KEY, CourseName VARCHAR(50), Fees DECIMAL(10,2) );
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0306 seconds.)

```
CREATE TABLE Enrollment ( EnrollmentID INT PRIMARY KEY, StudentID INT, CourseID INT, EnrollmentDate DATE, FOREIGN KEY (StudentID) REFERENCES Student(StudentID), FOREIGN KEY (CourseID) REFERENCES Course(CourseID) );
```

Console



Q2) Insert at least 6 records into the Student table, 4 records into the Course table, and 5 records into the Enrollment table.

Make sure:

Requirement
At least one student should not have enrollment
At least one course should not have any enrolled student

QUERY:

-- Insert into Student table

INSERT INTO Student VALUES

(1, 'Alice', 'NY'),

(2, 'Bob', 'LA'),

(3, 'Charlie', 'SF'),

(4, 'Dave', 'CHI'),

(5, 'Eve', 'MIA'),

(6, 'Frank', 'SEA');

-- Insert into Course table

INSERT INTO Course VALUES

(101, 'Math', 500.00),

(102, 'Science', 600.00),

(103, 'Art', 300.00),

(104, 'History', 400.00);

-- Insert into Enrollment table

INSERT INTO Enrollment VALUES

(1001, 1, 101, '2023-01-01'),

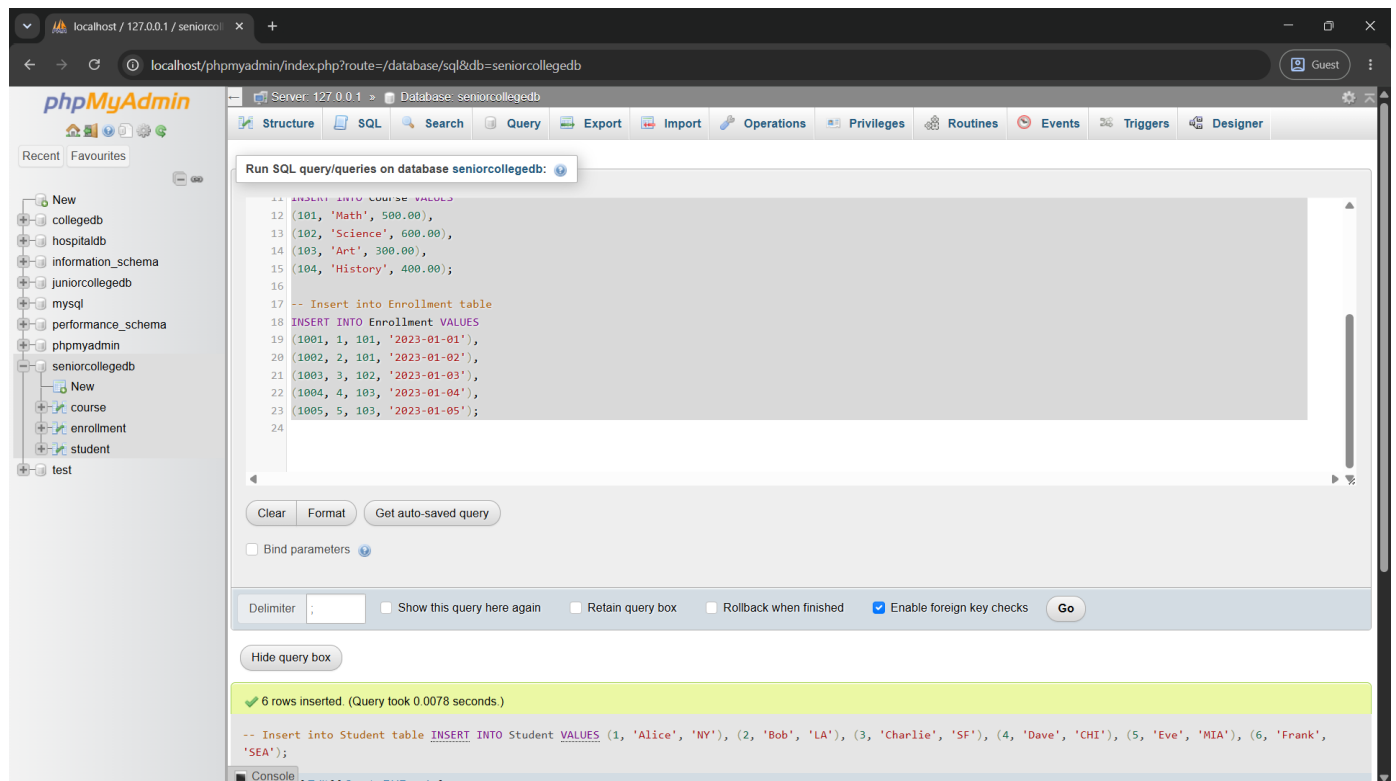
(1002, 2, 101, '2023-01-02'),

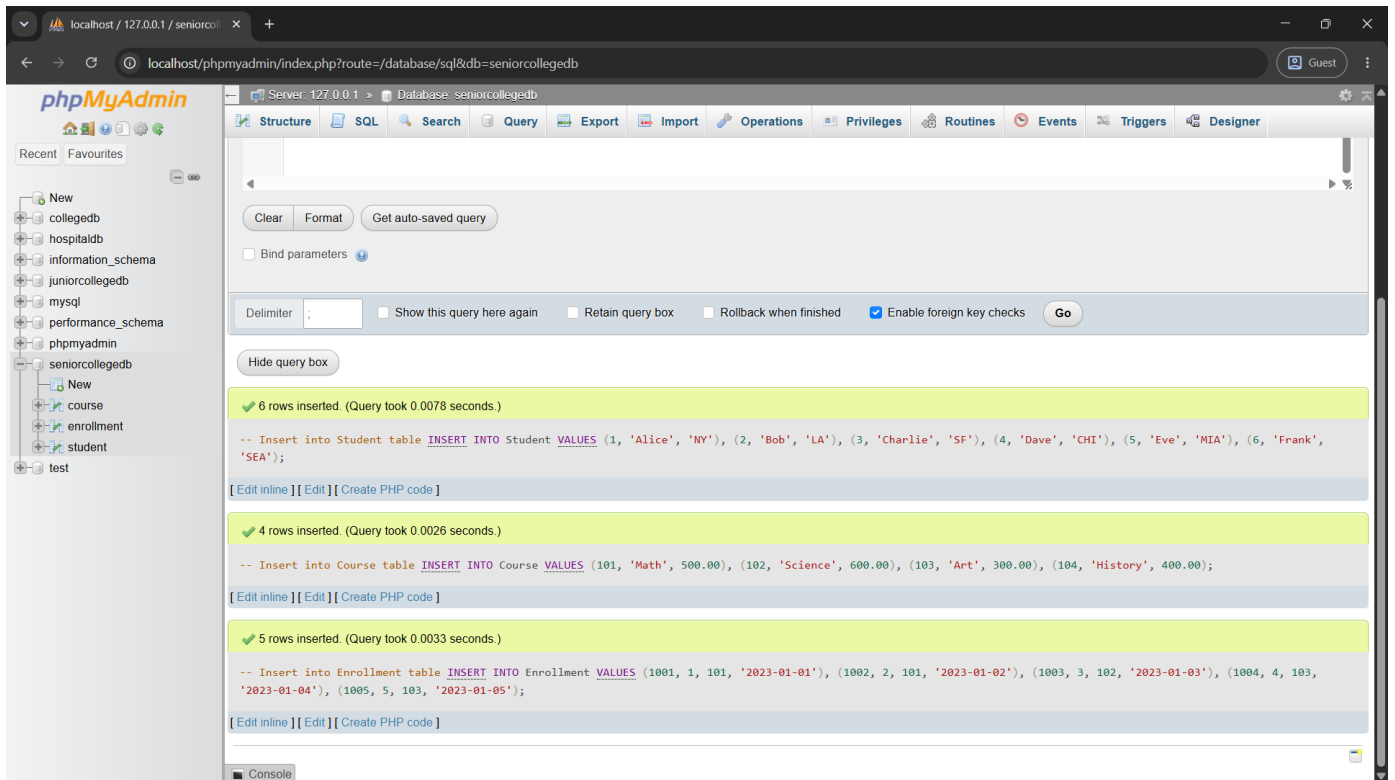
(1003, 3, 102, '2023-01-03'),

(1004, 4, 103, '2023-01-04'),

(1005, 5, 103, '2023-01-05');

OUTPUT:





Q3) Write a MySQL query using INNER JOIN to display students who have enrolled in courses.

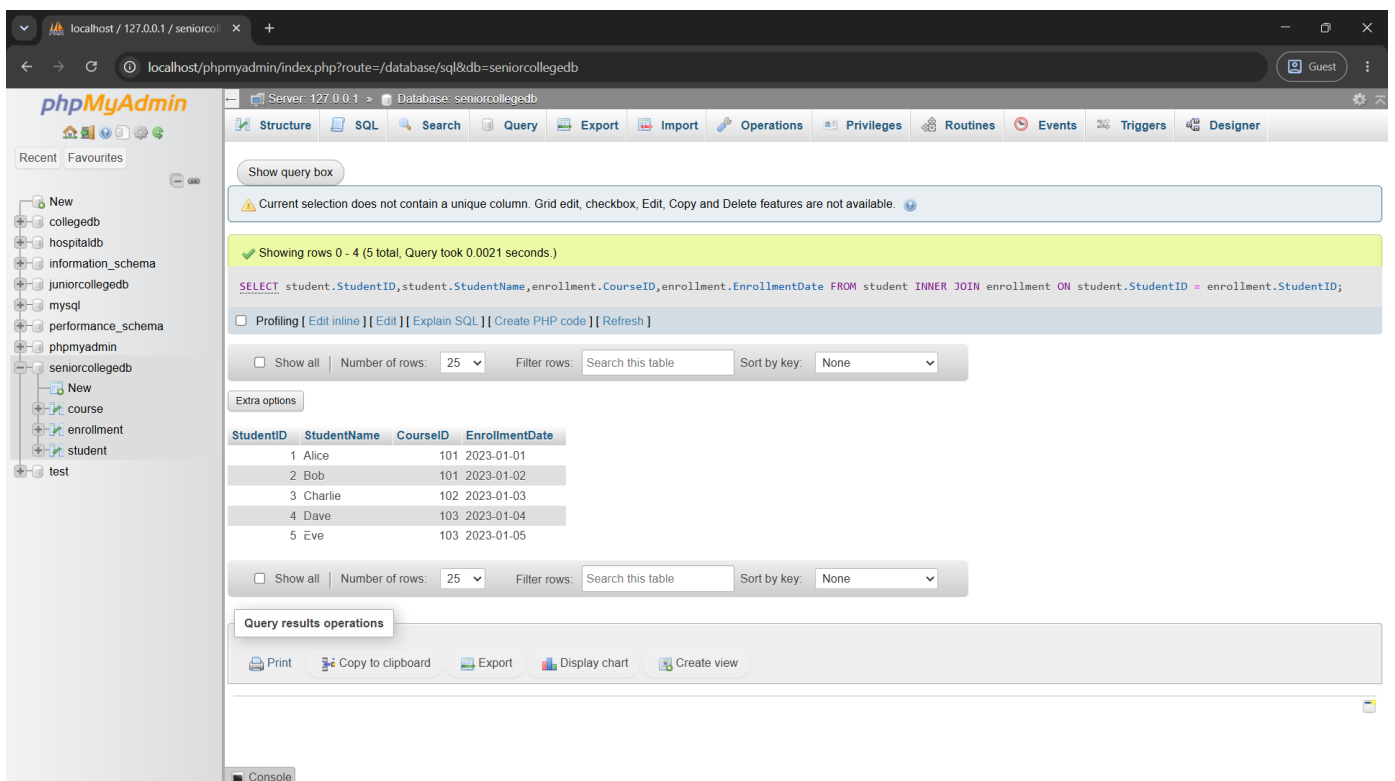
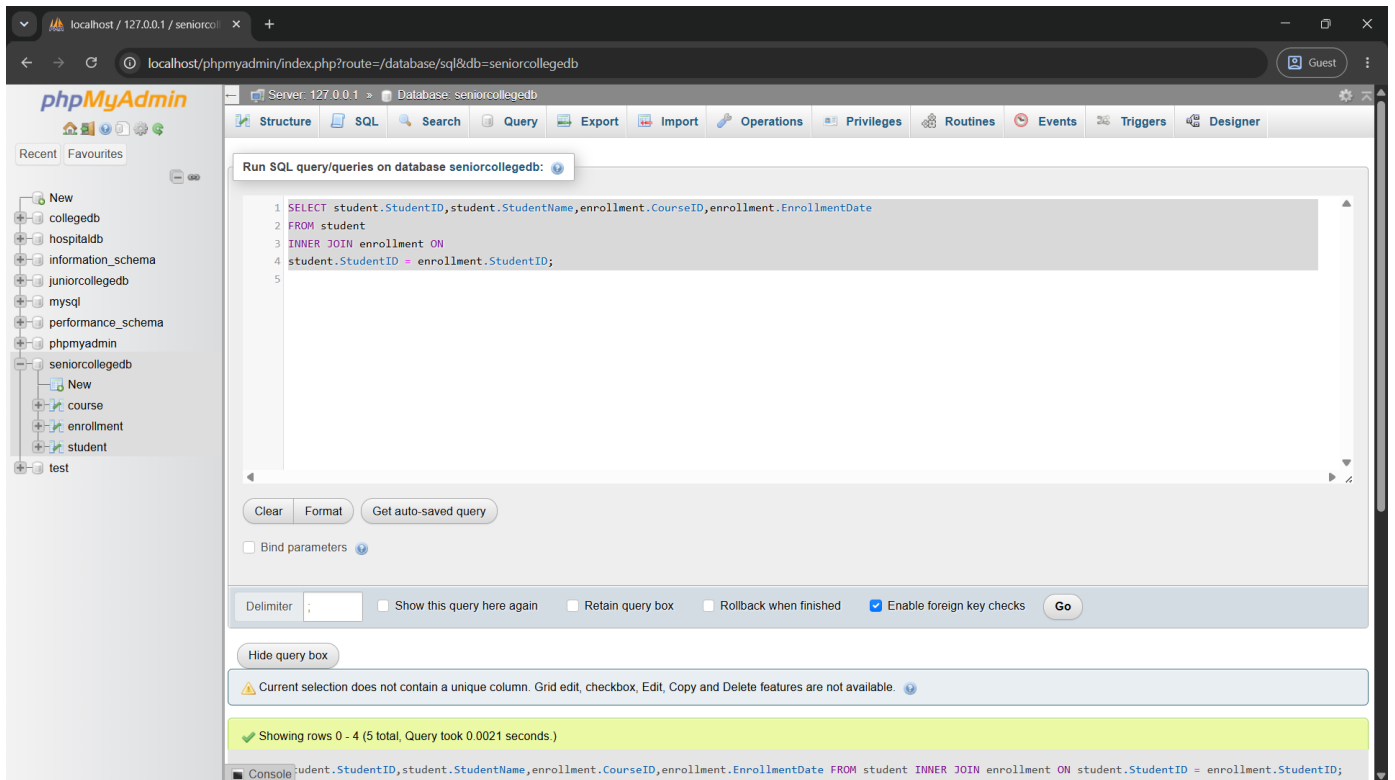
The output should include:

Column
StudentID
StudentName
CourseID
EnrollmentDate

QUERY:

```
SELECT student.StudentID,student.StudentName,enrollment.CourseID,enrollment.EnrollmentDate
FROM student
INNER JOIN enrollment ON
student.StudentID = enrollment.StudentID;
```

OUTPUT:



Q4) Write a MySQL query to display student name, city, course name, fees, and enrollment date using multiple tables.

Use:

Tables Required

Student
Course
Enrollment

QUERY:

SELECT

Student.StudentName,Student.City,Course.CourseName,Course.Fees,Enrollment.EnrollmentDate

FROM student

JOIN Enrollment ON

Student.StudentID = Enrollment.StudentID

JOIN Course ON Enrollment.CourseID = Course.CourseID;

OUTPUT:

The screenshot shows the phpMyAdmin interface for the 'seniorcollegedb' database. The SQL query is entered in the 'Run SQL query/queries on database seniorcollegedb:' box. The query is as follows:

```

1 SELECT Student.StudentName,Student.City,Course.CourseName,Course.Fees,Enrollment.EnrollmentDate
2 FROM student
3 JOIN Enrollment ON
4 Student.StudentID = Enrollment.StudentID
5 JOIN Course ON Enrollment.CourseID = Course.CourseID;
6

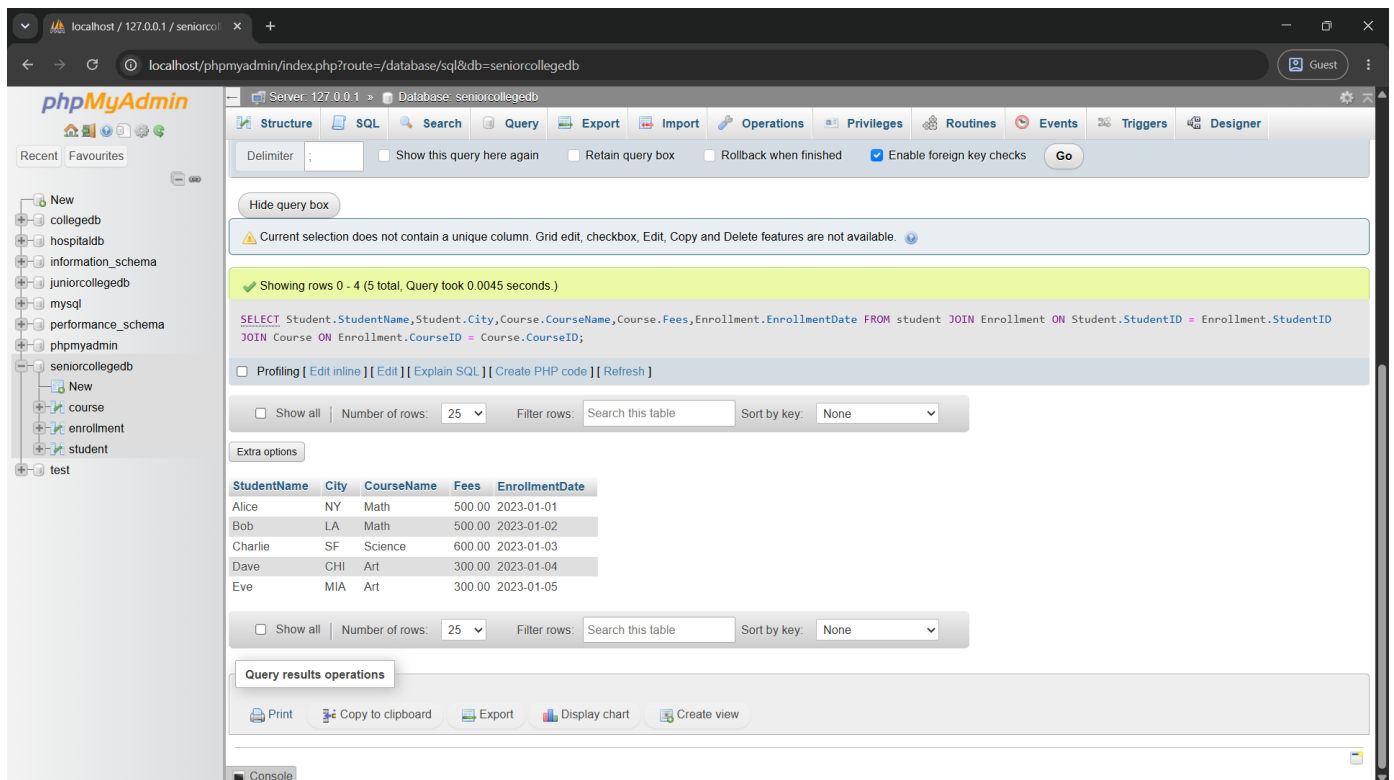
```

Below the query box, there are buttons for 'Clear', 'Format', and 'Get auto-saved query'. There is also a checkbox for 'Bind parameters' which is currently unchecked. At the bottom of the query box, there are options for 'Delimiter' (set to semicolon), 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (checked). A 'Go' button is also present.

Below the query box, there is a message: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.'

At the bottom, a green status bar indicates: 'Showing rows 0 - 4 (5 total, Query took 0.0045 seconds.)'

The console at the bottom shows the executed query: 'Student.StudentName,Student.City,Course.CourseName,Course.Fees,Enrollment.EnrollmentDate FROM student JOIN Enrollment ON Student.StudentID = Enrollment.StudentID'.



Q5) Write a MySQL query using LEFT JOIN to display all students, including those who have not enrolled in any course.

The output should include:

Column
StudentID
StudentName
City
CourseID
EnrollmentDate

QUERY:

```
SELECT Student.StudentID, Student.StudentName, Student.City, Enrollment.CourseID,
       Enrollment.EnrollmentDate
FROM Student
LEFT JOIN Enrollment
ON Student.StudentID = Enrollment.StudentID;
```

OUTPUT:

Run SQL query/queries on database seniorcollegedb:

```

1 SELECT Student.StudentID, Student.StudentName, Student.City, Enrollment.CourseID,
2       Enrollment.EnrollmentDate
3 FROM Student
4 LEFT JOIN Enrollment
5 ON Student.StudentID = Enrollment.StudentID;
6
7

```

Showing rows 0 - 5 (6 total, Query took 0.0002 seconds)

Console: Student.StudentID, Student.StudentName, Student.City, Enrollment.CourseID, Enrollment.EnrollmentDate FROM Student LEFT JOIN Enrollment ON Student.StudentID = Enrollment.StudentID;

Showing rows 0 - 5 (6 total, Query took 0.0002 seconds)

SELECT Student.StudentID, Student.StudentName, Student.City, Enrollment.CourseID, Enrollment.EnrollmentDate FROM Student LEFT JOIN Enrollment ON Student.StudentID = Enrollment.StudentID;

Extra options

StudentID	StudentName	City	CourseID	EnrollmentDate
1	Alice	NY	101	2023-01-01
2	Bob	LA	101	2023-01-02
3	Charlie	SF	102	2023-01-03
4	Dave	CHI	103	2023-01-04
5	Eve	MIA	103	2023-01-05
6	Frank	SEA	NULL	NULL

Query results operations

Print Copy to clipboard Export Display chart Create view

Q6) Write a MySQL query using RIGHT JOIN to display all courses, including courses that have no enrolled students.

The output should include:

Column
StudentName

CourseID
CourseName
Fees

QUERY:

SELECT Student.StudentName, Course.CourseID, Course.CourseName, Course.Fees

FROM Student

JOIN Enrollment ON

Student.StudentID = Enrollment.StudentID

RIGHT JOIN Course ON

Enrollment.CourseID = Course.CourseID;

OUTPUT:

The screenshot shows the phpMyAdmin web interface. The left sidebar displays the database structure, including 'seniorcollegedb' and its tables: 'course', 'enrollment', and 'student'. The main panel shows the 'SQL' tab with the following query:

```

1 SELECT Student.StudentName, Course.CourseID, Course.CourseName, Course.Fees
2 FROM Student
3 JOIN Enrollment ON
4     Student.StudentID = Enrollment.StudentID
5 RIGHT JOIN Course ON
6     Enrollment.CourseID = Course.CourseID;
7

```

Below the query editor, there are buttons for 'Clear', 'Format', and 'Get auto-saved query'. A checkbox for 'Bind parameters' is also present. The 'Go' button is highlighted. Below the query editor, there are checkboxes for 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (which is checked). A 'Hide query box' button is also visible.

A message bar indicates: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.'

A green message bar shows: 'Showing rows 0 - 5 (6 total, Query took 0.0003 seconds)'. Below this, the console displays the query results:

```

student.StudentName, Course.CourseID, Course.CourseName, Course.Fees FROM Student JOIN Enrollment ON Student.StudentID = Enrollment.StudentID RIGHT JOIN Course ON

```

localhost / 127.0.0.1 / seniorcol

localhost/phpmyadmin/index.php?route=/database/sql&db=seniorcollegedb

Guest

Server: 127.0.0.1 » Database: seniorcollegedb

StructureSQLSearchQueryExportImportOperationsPrivilegesRoutinesEventsTriggersDesigner

Hide query box

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 5 (6 total. Query took 0.0003 seconds)

SELECT Student.StudentName, Course.CourseID, Course.CourseName, Course.Fees FROM Student JOIN Enrollment ON Student.StudentID = Enrollment.StudentID RIGHT JOIN Course ON Enrollment.CourseID = Course.CourseID;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

StudentName	CourseID	CourseName	Fees
Alice	101	Math	500.00
Bob	101	Math	500.00
Charlie	102	Science	600.00
Dave	103	Art	300.00
Eve	103	Art	300.00
NULL	104	History	400.00

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

PrintCopy to clipboardExportDisplay chartCreate view

Console